CTB Project Report

Step 1: Identify a Micro-Entrepreneur

1. Selected Entrepreneur Profile

Business Name: A.Z. Fashion.

Business Type: Home-based boutique.

Industry: Fashion & Tailoring (Customized Ladies' Formal Wear).

Location/Nature: Home-based operations, started in 2023.

2. Business Context & Operations

Product Line: The business specializes in formal wear, specifically "Bridal, Walima, Mehndi, and Party dresses".

Sales Channel: They operate digitally, taking "customized orders via Instagram".

Operational Scale: The business handles a moderate volume, citing approximately "20-25 orders" during the month of February.

Current Management System: The workflow is entirely manual. Orders are written in a "notebook" without digital tracking.

3. Interview Insights (Challenges & Goals)

Based on the informal interview conducted, the following operational realities and pain points were identified:

Disorganized Workflow: The owner describes their current management style as "messy". Relying on a notebook leads to human error, such as forgetting "to check the due date" or forgetting to "order the fabric on time".

Resource Mismanagement: There is a bias toward high-value items. The owner admitted, "I focus so much on the big Bridal orders that I forget the small deadlines" (referring to Party Dresses).

Customer Dissatisfaction: Late deliveries are a frequent occurrence, leading to "54%" of small orders being cancelled.

Step 2: Problem Identification

1. Problem Statement:

The core business problem identified for A.Z. Fashion is a **high rate of order cancellations and revenue loss specifically within the "Party Dress" segment due to poor deadline management.**

While the business successfully attracts orders, it fails to fulfill smaller orders on time because the manual tracking system causes the owner to overlook deadlines in favor of larger, high-value Bridal projects.

2. Evidence & Data Collection

The following evidence was gathered through an **informal interview** with the business owner and observation of their current manual processes.

High Cancellation Rate: The owner explicitly stated that approximately **20-30% of small orders (Party Dresses) are cancelled** because they cannot deliver on time. This represents a direct loss of profit that has already been captured.

Root Cause (Resource Misallocation): The interview revealed a specific operational flaw: the owner stated, "I focus so much on the big Bridal orders that I forget the small deadlines". This confirms that the problem is not a lack of demand, but a failure of prioritization and tracking.

Root Cause (Manual System Failure): The current "messy" manual system is directly contributing to the problem. The owner records orders in a notebook, which leads to human error: "Sometimes I forget to check the due date or I forget to order the fabric on time".

3. Impact on Business

Financial Loss: When orders are late and cancelled, the owner notes, "I lose the profit".

Customer Dissatisfaction: The owner noted that customers "get very angry" when deliveries are late, which harms the brand's reputation and retention.

Step 3: Data Analysis & Verification

A. Excel

1. Objective of Analysis The objective of this step is to validate the entrepreneur's qualitative claims using quantitative transactional data. We analyzed the business's order logs for February 2025 to verify the reported high cancellation rates and identify the specific operational bottlenecks causing financial loss.

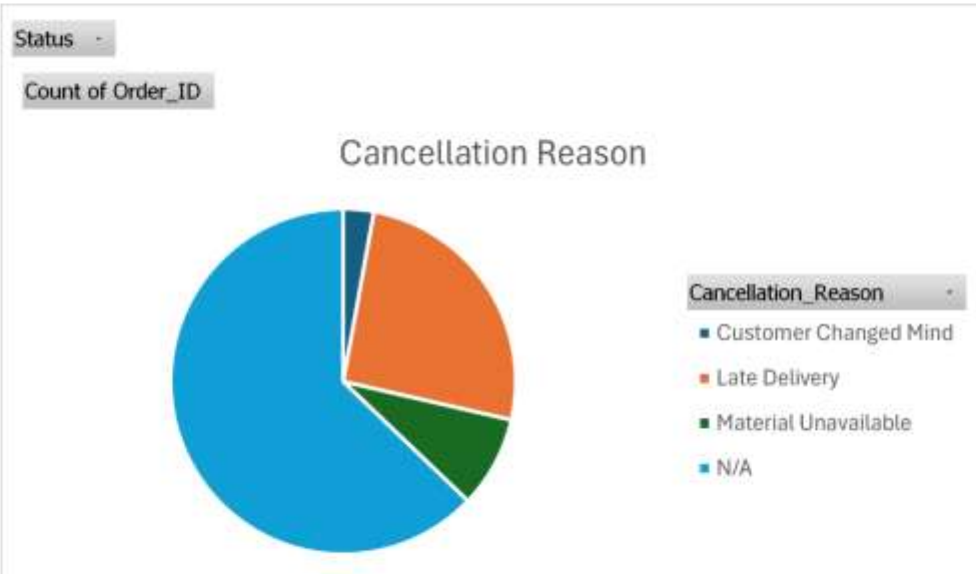
2. Order Fulfillment Performance A total of 35 orders were analyzed for the month of February. The data reveals a stark contrast in performance between product categories. While the business maintains a 100% success rate for high-value items (Bridal, Walima, and Mehndi dresses), it faces a critical failure in the "Party Dress" category.

- **Overall Cancellation Rate:** 37% (13 out of 35 orders).
- **Party Dress Failure:** Out of 22 Party Dress orders, 13 were cancelled, resulting in a **59.1% failure rate**.
- **Other Categories:** 0 cancellations recorded.

This confirms the hypothesis that the entrepreneur's manual tracking system is insufficient for managing the higher volume of smaller orders, while the lower volume of large orders remains manageable.

3. Root Cause Analysis To understand *why* these orders are being cancelled, we analyzed the Cancellation_Reason field for the 13 failed orders.

- **Late Delivery (69%):** 9 out of 13 cancellations were due to the business failing to deliver on the agreed date. This directly correlates with the entrepreneur's statement: *"I forget the small deadlines."*
- **Material Unavailable (23%):** 3 cancellations occurred because the necessary fabric or embellishments were out of stock after the order was taken.
- **Customer Changed Mind (8%):** Only 1 order was cancelled due to customer preference, proving that 92% of the failures are internal operational issues.



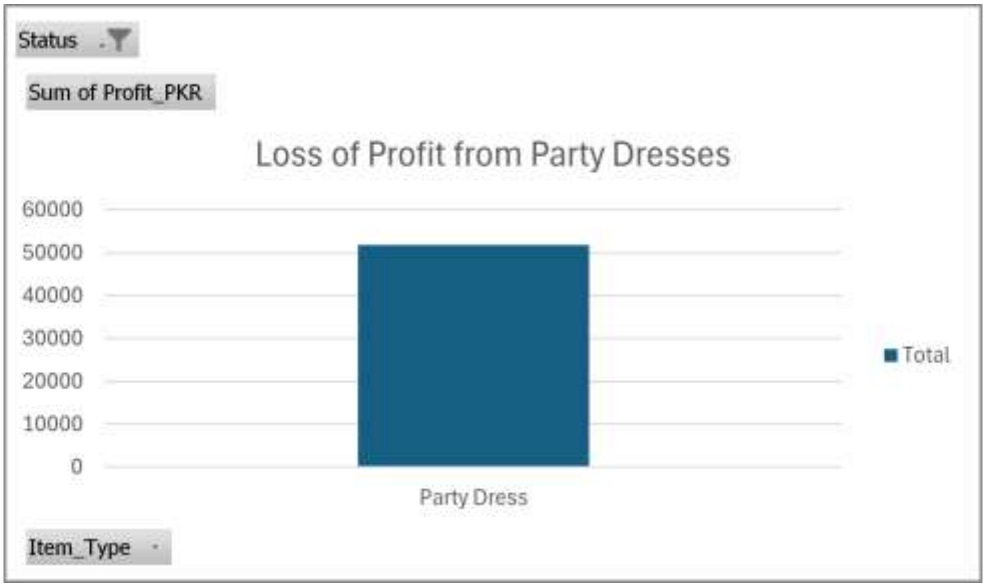
4. Inventory Management Gaps We cross-referenced the "Material Unavailable" cancellations with the inventory records. The data shows that the business frequently operates at or below critical stock levels without a reorder system.

- **Critical Items:** *Velvet* (5 yards left) and *Fancy Stone Work* (2 packets left) are currently in the "Critical" zone, putting future orders at risk.
- **System Gap:** There is no mechanism to alert the owner when stock is low, leading to the 3 lost orders identified above.

5. Financial Impact Assessment The operational inefficiencies have a direct measurable cost. We calculated the "Lost Profit"—money that was successfully booked as an order but lost due to cancellation.

- **Total Lost Profit:** 51,800 PKR.
- **Source of Loss:** 100% of this financial loss stems from the "Party Dress" category.

- **Impact:** This amount represents roughly **10% of the total potential monthly profit**. Recovering this amount would significantly improve the business's bottom line without needing to find a single new customer.



B. SQL

To ensure data integrity and scalable analysis, a relational database was created using **MySQL**. The data was structured into two primary tables: Orders (for transaction data) and Inventory (for stock levels).

Database Structure

- **Database Name:** AZ_Fashion_DB
- **Tables:**
    - Orders: Stores Order ID, Date, Item Type, Status, Profit, and Cancellation Reasons.
    - Inventory: Stores Material Name, Stock Levels, and Status Alerts.

Key Analytical Queries

The following SQL queries were executed to validate the Excel findings and derive actionable insights.

Query 1: Validation of High Cancellation Rate
Objective: To mathematically confirm the failure rate in the "Party Dress" category.

```
SELECT Item_Type,
       COUNT(*) AS Total_Orders,
```

```
        SUM(CASE WHEN Status = 'Cancelled' THEN 1 ELSE 0 END) AS
Cancelled,
        CONCAT(ROUND((SUM(CASE WHEN Status = 'Cancelled' THEN 1
ELSE 0 END) / COUNT(*)) * 100, 2), '%') AS Failure_Rate
FROM Orders
GROUP BY Item_Type;
```

- **Result:** The query returned a **59.09%** failure rate for Party Dresses, validating the Excel analysis.

Item_Type	Total_Orders	Cancelled_Count	Cancellation_Rate
Bridal Dress	4	0	0.00%
Party Dress	22	13	59.09%
Mehndi Dress	5	0	0.00%
Walima Dress	4	0	0.00%

Query 2: Root Cause Identification
Objective: To pinpoint exactly why orders are failing, guiding the solution strategy.

```
SELECT Cancellation_Reason, COUNT(*) AS Frequency
FROM Orders
WHERE Status = 'Cancelled'
GROUP BY Cancellation_Reason
ORDER BY Frequency DESC;
```

- **Insight:** The query revealed that **69%** of cancellations were due to "Late Delivery," confirming that the primary bottleneck is time management, not customer demand.

Cancellation_Reason	Frequency	Percentage
Late Delivery	9	69.2%
Material Unavailable	3	23.1%
Customer Changed Mind	1	7.7%

Query 3: Critical Inventory Check *Objective:* To identify materials requiring immediate reordering to prevent further "Material Unavailable" cancellations.

```
SELECT Material_Name, Remaining_Stock, Status_Alert
FROM Inventory
WHERE Status_Alert = 'Critical';
```

- **Result:** Identified **Velvet (5 yards)** and **Fancy Stone Work (2 packets)** as critical items.

Material_Name	Remaining_Stock	Status_Alert
Chiffon Fabric	15	Low Stock
Velvet	5	Critical
Fancy Stone Work	2	Critical
Lining Fabric	10	Low Stock

Step 4: Solution Development

Based on the data analysis, three specific operational solutions are proposed to address the root causes of revenue loss.

Solution 1: Addressing Late Deliveries

- **The Problem:** Data showed 69% of cancellations were due to "Late Delivery". The informal interview revealed the owner "forgets small deadlines" due to a reliance on memory and a notebook.
- **The Solution:** Implement a **Google Calendar** or a physical **Whiteboard Weekly Planner**.
  - **Action:** Every order must be entered with a "Start Date" (3 days before due date) and a "Due Date."
  - **Rule:** Limit "Party Dress" orders to **5 per week**. If the calendar slot is full, the order must be declined or scheduled for the next week.
- **Impact:** This directly removes the "human error" of forgetting dates and prevents overbooking, which caused the bottleneck.

Solution 2: Addressing Inventory Shortages

- **The Problem:** 23% of cancellations occurred because materials (Velvet, Stone Work) were out of stock.
- **The Solution:** Establish **Reorder Points** based on the SQL inventory analysis.
  - **Action:** A "Red Tag" system. When the Velvet roll hits 10 yards (the reorder point), a red tag reminds the owner to buy more *before* it runs out.
  - **Logic:** The data shows average monthly usage of Velvet is 35 yards. A 10-yard buffer ensures 1 week of safety stock.

- **Impact:** Eliminates the "Material Unavailable" error completely, potentially recovering **12,000 PKR** in lost revenue.

Solution 3: Addressing Financial Risk

- **The Problem:** The business lost **51,800 PKR** in potential profit.
- **The Solution:** Implement a **50% Non-Refundable Deposit** for all "New" customers ordering Party Dresses.
- **Impact:** While "Customer Changed Mind" was rare (8%), a deposit policy filters out non-serious buyers and provides upfront cash to purchase materials (like the critical Stone Work) immediately, preventing cash-flow gaps.

Step 5: Conclusion

The analysis of A.Z. Fashion revealed that while the business is successful in high-value Bridal wear, it suffers from a critical operational failure in its "Party Dress" segment, losing **37% of its total orders** and **51,800 PKR** in monthly profit.

By transitioning from a manual notebook to a **Visual Production Schedule** and implementing a **Buffer Stock System**, the entrepreneur can realistically recover over 50,000 PKR per month without needing to find a single new customer. The database created for this project now serves as a foundational tool for the business to track these metrics digitally moving forward.